

Comments and Corrections

Corrections to: “Green Networking With Packet Processing Engines: Modeling and Optimization”

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THE purpose of this note is to provide corrections to some typos that appeared in [1], as well as to point out the correct expression of the average waiting time for the $M^X/D/1/SET$ queueing system used therein in the derivation of the optimization results.

Specifically, according to the original derivation from [2], [1, p. 115, eq. (19)] should read as follows:

$$\begin{aligned}\bar{L} &= \lim_{z \rightarrow 1} P'_s(z) - \rho \\ &= \frac{2\lambda\beta\tau_s + \lambda^2\beta^2\tau_s^2 + \lambda\tau_s(\sum_{j=1}^{j_{max}} \beta_j j^2 - \beta)}{2(1 + \lambda\beta\tau_s)} \\ &\quad + \frac{\rho^2 + \frac{\lambda}{\mu}(\sum_{j=1}^{j_{max}} \beta_j j^2 - \beta)}{2(1 - \rho)}\end{aligned}\quad (1)$$

with $\bar{L}$ representing the mean number of packets in queue (and not in the system, as indicated in the sentence above Eq. (19)). Consequently, Eq. (20) for the average waiting time should become:

$$\bar{W} = \frac{\bar{L}}{\lambda\beta} = \frac{2\beta\tau_s + \lambda\beta^2\tau_s^2 + \tau_s(\sum_{j=1}^{j_{max}} \beta_j j^2 - \beta)}{2\beta(1 + \lambda\beta\tau_s)} + \frac{\rho\beta + (\sum_{j=1}^{j_{max}} \beta_j j^2 - \beta)}{2\mu\beta(1 - \rho)}\quad (2)$$

However, it must be also noted that the expression of the Probability Generating Function (PGF) in [1, eq. (10)], which was derived from [2] (and, consequently, the expression of $P_s(z)$ in [1, eq. (12)], from which (1) and (2) above derive), is incorrect. The correct general expression can be found in [3] (Eq. (4), p. 1285); specializing it to our case of deterministic service and setup times results in

$$\zeta(z) = \frac{1 - X(z)e^{-\lambda\tau_s[1-X(z)]}}{(1 + \lambda\tau_s)[1 - X(z)]}\quad (3)$$

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and then

$$P_s(z) = \frac{1 - X(z)e^{-\lambda\tau_s[1-X(z)]}}{(1 + \lambda\tau_s)[1 - X(z)]} \frac{(1 - \rho)(1 - z)e^{-\frac{\lambda}{\mu}[1-X(z)]}}{e^{-\frac{\lambda}{\mu}[1-X(z)]} - z}\quad (4)$$

Therefore, the final correct expressions of $\bar{L}$ and $\bar{W}$ in our case are given by

$$\bar{L} = \frac{\lambda\beta(2\tau_s + \lambda\tau_s^2)}{2(1 + \lambda\tau_s)} + \frac{\rho^2 + \frac{\lambda}{\mu}(\sum_{j=1}^{j_{max}} \beta_j j^2 - \beta)}{2(1 - \rho)}\quad (5)$$

$$\bar{W} = \frac{2\tau_s + \lambda\tau_s^2}{2(1 + \lambda\tau_s)} + \frac{\rho\beta + (\sum_{j=1}^{j_{max}} \beta_j j^2 - \beta)}{2\mu\beta(1 - \rho)}\quad (6)$$

Eq. (6) is consistent with the result in [4], where it is derived by starting from the PGF of the waiting time. Based on the derivations in [5, p. 299], a further modification is also necessary in [1, eqs. (17), (18), and (23)]: specifically, the factor β disappears from the numerator in the second term of Eq. (17) and in (18), and from all the denominators in (23).

As regards the numerical results in [1], they are substantially unaffected by the substitution of Eq. (2) with Eq. (6), as well as by the other modifications noted above, owing to the relatively limited burstiness of the traffic traces used to derive them. Significant variations would appear only in the presence of highly bursty traffic. The same observation applies for the results in [6], where the part of the expression of the average waiting time stemming from the presence of the setup time should also be modified according to the results of [3] and [4].

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